NATIONAL UNIVERSITY



Fourth Year Syllabus
Department of Environment Science

Four Year B.Sc. Honours Course
Effective from the Session: 2013–2014

National University

Subject: Environment Science
Syllabus for Four Year B.Sc. Honours Course
Effective from the Session: 2013-2014

Year wise Papers and marks distribution

FOURTH YEAR

Paper Code	Paper Title	Marks	Credits
244401	Research Methodology	100	4
244403	Climate Change	100	4
244405	Environmental Hazards and Disaster Managements	100	4
244407	Water Resources Management in Bangladesh	100	4
244409	Waste Management	100	4
244411	Environmental Laws, Protocols and Ethics	100	4
244413	Instrumental Techniques in Environmental Analysis	100	4
244415	**Energy, Environment and Sustainable Development: Bangladesh Perspective**	100	4
244417	Agriculture and Environment	100	4
244418	Viva-voce	100	4
	Total =	1000	40

Detailed Syllabus

Paper Code	244401	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	Research Methodology			

1. Introduction: Objectives and importance of research methodology; Approach: Methodology and limitation of research; Research design: features of good design, concept and development of research plan: Sampling design – sampling strategy, methodology and types of design of the program, census and sample survey: Measurement and scaling techniques.
2. Research Planning Fundamental and Processes; An Overview of Survey Methodology; Developing goals and objectives; Indicator analysis: Operational Plan Development; Methodology, tools and techniques for data collection; Resource Planning; Planning Under Risk and Uncertainty; Need Assessment; Concept Paper Development and Proposal Writing; Writing Evaluation Report.
3. Conceptualization of research problems; Research design, techniques of sampling design, types of sampling procedure experimental designing with an specific ecosystem; Concepts of Research plan, Practices for developing comprehensive research plan. Principles of presenting research finding and writing scientific papers and reports. Criteria for selecting suitable means of presentation and their explanation.
4. Methods of data collection – primary data, secondary data, selection of appropriate data, guidelines for interviewing, questionnaire and schedule: Social and participatory methodologies in peoples' participation: Data processing and analysis – problems, types, statistics, interpretation of results, graphical representation and tabulation, manuscript preparation, rules of quotation and bibliography: Report writing: Case studies.

Recommended References:

1. Bishop, (1992) ON.19. Statistics for Biology, A practical guides for the experimental Biologists, Longman,, UK.
2. Blalock, H.M.Jr. (1979): Social Statistics, MacGraw Hill Book Company, New Delhi.
3. Cochran, W. G. and Cox, G. M. (1957): Experimental Designs; John Wiley and Sons nc, London.
4. Cochran, W. G. (1963). Sampling Techniques (second edition), John Wiley & Sons, NY
5. Islam, M. Nurul. (2008). An Introduction to Sampling Methods. Mullick and Brothers. New Market, Dhaka.
6. Islam, M. Nurul. (2009). An Introduction to Research Methods. Mullick and Brothers. New Market, Dhaka.
7. Wilson: Introduction to Research Methodology. McGraw-Hill Book Co, NY.

Paper Code	244403	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	Climate Changes			

1. **The climate System:** Climate and the climate system, Natural forcing of the climate system, Natural variability of climate, Extreme events, human influence on the climate system, modelling and projection of anthropogenic climate change, observing anthropogenic climate change.
2. **Observed climate variability and change:** Land-surface, sea surface temperature, sea air temperature, Temperature of the upper air, changes in the cryosphere. Temperature of the past

1000 years, volcanic and solar effects, changes in precipitation and related variable (Land, Palaeo-drought, Ocean), water vapor, evaporation, clouds the atmospheric / oceanic circulation change, extreme weather and climate events.

3. **The carbon cycle and Atmospheric carbon dioxide:** Terrestrial and ocean biogeochemical carbon cycle, Palaeo CO2 and Natural changes in the carbon cycle, anthropogic source of CO2, CO2 concentration measurement methods, Carbon cycle model evaluation, Projection of CO2 concentration and their implications.
4. **Green house gases and their Impact:** Trace gases, current observation, trends, projection of future emissions, overall impact of global atmospheric chemistry change.
5. **Radiative Forcing of climate change:** Radiative forcing, forcing-response relationship, well-mixed greenhouse gases, stratospheric ozone, land-use change, solar forcing of climate, global warming potentials, Global mean radiative forcing, Time evolution of radiative forcing.
6. **Model Evaluation:** Mean state and variability in climate models, model Hierarchy, coupled climate models-some methodologies, coupled climate models-means, 20th century climate and climate variability, coupled model Phenomena, extreme events, coupled models-dependence on resolution, sources of uncertainty and levels of confidence in coupled models.
7. **Projections of future climate change:** Climate and climate change, global mean response, patterns of future climate change, range of temperature response, factors that contribute of the response, changes in variability, charges of extreme events.
8. **Changes in Sea level:** Factors contributing to sea level change, Past sea level changes, future sea level changes, reducing the uncertainties in future estimates of sea level change.
9. **Mitigation options:** Energy supply mitigation options, Industry, transportation sector, human settlements, agricultural options, management of forests, Reducing Emissions from Deforestation and Forest Degradation in Developing Countries (*REDD*), Carbon sequestration, trading and Taxing.
10. **Climate Change and Bangladesh:** Scenarios, Impacts and adaptation measures.
11. **Climate Change and health:** Health and environmental effects, Climate sensitive diseases, Climate induced natural hazards and health, Uncertainties.

Recommended References:

1. Climate change 2007 by, J.T. Houghton, Y. Ding, D.J. Griggs, Cambridge University press 2007.
2. Climate Process and Change by Edward Bryant, Cambridge University Press, 1997.
3. Bangladesh: Climate Change Impacts and Vulnerability: A Synthesis by Ahsan Uddin Amhed, 2006., Climate Change Cell, Department of Environment, Bangladesh.
4. Documents published by Climate Change Cell, Department of Environment, Bangladesh (2009).

Paper Code	244405	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Environmental Hazards and Disaster Managements**			

1. **Risk Assessment:** Concept of risk, hazard, disaster and vulnerability, factors of risk, types of risk, point of risk analysis, risk management goals, strategies, principles of risk management framework, risk management methods, principles of decision making, and public perception of risk.

2. Geoenvironmental problems identification and risk management, framework for risk-based site management, defining goals, regulatory and societal issues, site assessment for risk-based site management.
3. **Hazard in the environment:** definition of event, hazard and disaster, common feature of disaster, classification of disaster, factor related to damage, potential hazardous process or agencies, hazard caused stages and sequences, dimension of disaster.
4. Formation process, exposure and impact of various national hazard: Cyclones, storm surges, thunderstorms, (kalboishakhi), lightening, tornadoes, flooding, drought, landslides, subsidence, tsunamis, Volcanic activity.
5. **Earthquake engineering:** seismological aspects, earthquake occurrences, wave type, earthquake stations, source zones, magnitude, probability of occurrence, mode of wave propagation, designed ground motions, seismic soil structure interactions, and seismic design of retaining structures, liquefaction, and seismic risk in geotechnical earthquake engineering.
6. Coastal hazard and disaster Management: **Types and causes of coastal hazard, Adjustment of hazard, Warning and forecasting system, case study in Bangladesh.**
7. **Hazard and risk assessment:** technique, uncertainty, warning system, types and levels, natural resources maps, hazard related maps, use of hazard information in the development and planning process, stage of development planning, integrated flood hazard assessment in to development planning process, Community risk assessment.
8. **Disaster and hazard reduction:** Scope of mitigation of environmental disaster and hazards, methods of mitigation, livelihood adaptation, risks reduction model, institution, Comprehensive disaster management programme, disaster management model, disaster management policy in Bangladesh, early warning information and GIS based disaster technology.

Recommended References:

1. Smith, 1995, Environmental Hazard, 2nd edition.
2. Edward A. Keller, 1985, 4th edition. Environmental Geology. CBS Publishers & Distributors, Delhi.
3. R. Kerry Rowe .2001, Geotechnical and Geoenvironmental Engineering Hand Book. Kluwer Academic Publishers, London.
4. Roy E. Hunt, Geotechnical Engineering analysis and Evaluation.
5. Applied Geology for Engineers, by: Military Engineering volume XV.
6. P. Leonard Capper, W. Fisher Cassie, and JD. Geddes, Problems in engineering soils,
7. Roy E. Hunt, geotechnical engineering investigation manual.
8. Dasgupta, Samir. 2009. *Understanding the Global Environment,* 2009. Delhi: Pearson-Longman.

Paper Code	244407	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Water Resources Management in Bangladesh**			

1. **Basic concepts in water resources management:** Definition, Water Resources System, Water Management principles, Basic concepts of Precipitation, evaporation, transpiration, infiltration, IWRM concepts, Overview of Water Resources Management in Bangladesh.

2. **Wetlands and water resources:** Soil-water relationship: Human impacts on water resources: Irrigation and flood control system, Economic, environment and institutional aspects of water resources management: Participation of beneficiaries, formation of user's group: Strategic planning: System analysis approach: Conceptual framework and models: Analytical techniques: Operation and maintenance of water resource system.
3. **Water Resources of Bangladesh:** Confluence of Rivers, Surface Water, Ground Water, Coastal Water, Rain, Water, Abstraction, Recharge, Regenerated Water.
4. **River Systems and Watershed Basins:** The Bengal Plain, River Systems, The Ganga River System, The Brahmaputra River System, The Meghna River System, The South-Eastern Rivers system; The Bengal Basin, Bangladesh, The North-West Region, The South-West Region, The North-Central Region, The South-Central Region, The North- East Region, The Central-Eas Region, The South-East Region, The Coastal Region.
5. **Climate & Floods:** The Climate, The Seasons, Rainfall, Evaporation, Flood Plains, Flood Basins, Flood Flows, Historical Floods, Flood Action Plan, Flood Management, Low Flows, Historical Cyclones, Cyclone Management, Threat of Sea Level Rise, The El Nino Effects.
6. **Soil & Agriculture:** Physiography, Geology, Topography, The Topsoil, Land Use, Flood Protection Projects, Submergible Embankments, The Crop Calendar, Crop Production, Cropping Pattern, Rice Production, Fish Production, Forest Resources.
7. **River Trainings and Navigation Routes:** Introduction, Erosion Types, Eroding Rivers, The Charlands, River Regime, Land Laws & Rivers, Erosion Control, River Flow Training, River Management, Seaside Protections, Harbors & Ports, City Protections, Materials; Types of Rivers, River Morphology, Perennial Rivers, Navigation Rivers, Navigation Khals, Historical Cuts, Navigation Routes, River Ferries, River Ports, Dredging, Dockyards, Port Facility, River Traveling, Aquatic Health.
8. **Haors & Wetlands:** Flood Plains, Low Lying Areas, Notional Importance, Seasonal Variations, Seasonal Wetlands, Wetlands Soil, Wetland Vegetations, Wetlands Birds, Wetland Crops, Wetlands Conservation, Haor Embankments, Drainage Projects, Water Conservation Projects, Oxbow Lakes, Wetland Management, Wildlife Sanctuaries.
9. **Water & Human Life:** Introduction, Rural Life, Urban Life, Metropolitan Life, Boat Life, Coastal Life, Hill Life, Char Life, Water Rights, Constraints, Human Activities, Natural Calamities, National Water Policy, Rural Water Supply, Urban Water Supply, Safety Standards, Dhaka Water Supply, Chittagong Water Supply, Sanitation, Urban Sanitation, DND Project, Disaster Threats, Dead or Dying Rivers, Reuse of Water, Recreation, Water Problems, Water footprint.
10. **Water & Industry:** Industry, Water Use, Virtual Water, Industrial Effluents, Dhaka Industries, Chittagong Industries, Cross-Boundary Pollution, National Water Policy, Environment Protection Act, Environment Court Act, City Public Lands Protection Act, Environmental Clearance, Water Standards, Pollution Control, Mining Control.
11. **Water & Economics:** Introduction, Water Projects, Water Economics, Project Implementation, Irrigation Projects, The Ganga Barrage Project, Small Scale Projects, Management Conflicts, Beneficiary Participation, Cost Recovery, Water Rates, O&M of Projects, Integrated Water Resources Management.
12. **Laws & Institutions:** Water Rights, Water Conflicts, Water Ethics, Rules of Business, Government Policies, Water Laws, Water Organizations, Water Forums, Water Conventions, Trans-Boundary Rivers, Water Treaties, India's Rivers Interlink Project, Domestic Water Disputes.
13. Case studies.

Recommended References:

1. Inamul Haque (2008) Water Resources Management in Bangladesh, Anushilan, Dhaka.
2. Pereira HC (1989) Policy and Practice in the Management of Tropical Watershed. Belhaven Press, London
3. Troch FR , Hobbs JA and Donahue RL (1991) Soil and Water Conservation. Prentice Hall, NJ
4. Ministry of Water Resources (1999) National Water Policy, Government of the People's Republic of Bangladesh
5. EGIS (2001) Guidebook for Integrated Water Resources Management: Concepts and Tools, Dhaka
6. Water Resources Planning Organisation (WARPO) (2001) Draft Final National Water Management Plan, Vol-1 & 2, WARPO, Dhaka.
7. Loucks, Stedinger and Haith, (1993): Water Resources System Planning & Analysis, McGraw Hill, USA.
8. Kabir, M. R. and Bhuiyan, A. B. M. F., (1997): Integrated Water Resources Management, WRE, BUET.
9. More reading material will be supplied at classroom.

Paper Code	244409	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Waste Management**			

1. **Introduction:** definition, classification of waste, general aspect of waste managements
2. **Solid waste management:** Characteristic of solid waste, solid waste collection, solid waste collection estimate and cost, waste transfer, maximum and economic haul time, disposal techniques, modes of operation, environmental consideration, liquid leachability, utilization of waste to energy production, resource conservation and recovery, low, high and medium technology.
3. **Hazardous waste management:** Types of hazardous waste, risk perception and risk assessment, toxicity consideration, one-hit mode, reasonable maximum exposure, risk characterization and risk management, transport regulations, disposal protocols.
4. **Radioactive Waste Management:** Definition, sources and characteristics of radioactive wastes, effects of radiation on living organism, acute effect and delayed effect, risk estimates, radioactive waste management techniques, safe disposal, long term assessment, case study.
5. **Treatment technologies of solid and hazardous wastes:** Biological, physical and chemical treatment, batch distillation and fractionation techniques, ion-exchange techniques, electrodialysis, reverse osmosis, waste incineration, land disposal and ground impounding.
6. **Municipal and household wastes management:** Disposal design, recovery and recycling of household wastes, domestic garbage, bulk treatment on commercial scale, recycling of paper.
7. **Hazardous waste and risk analysis:** Hazard identification, mutagenesis, carcinogenesis, dose-response relationship, human exposure assessment, risk characterization, contaminant degradation, application of risk assessment.
8. **Hazardous waste regulatory act:** The Comprehensive Environmental Response, Consumption and Liability Act (CERCLA), the National Priority List (NPL), the Hazard Ranking System (HRS), the National Contingency Plan (NCP), Superfund Amendments and Reauthorization Act (SARA), national and regional provisions of hazard treatment.

Recommended References:

1. Introduction to Environmental Engineering, M. L.. Davis, D. A. Cornwell, 3rd edn, McGrawHill, 1998.
2. Environmental Chemistry, C. Baird, WH Freeman and Company, N. Y., 2nd edn, 2000.
3. Environmental Chemistry, S. E. Manahan, 7th end, Lewis Publishers, 2000.
4. Introduction to Environmental Engineering and Science, G. M. Masters, Prentice-Hall Inc., 1991.

Paper Code	244411	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	Environmental Laws, Protocols and Ethics			

1. Definition of law, protocols, treaties and convention; classification of laws, environmental law, environmental rights, environmental justice, historical evolution of environmental legislation.
2. Mention the major UN conferences regarding environment: Stockholm declaration, 1972, Agenda 21, Rio-declaration, Kyoto protocol, CDM and Carbon credit, Convention of biological diversity, Ramser convention, Basel convention, UN convention on the laws of the sea; and other recent conventions.
3. Environmental law regime in Bangladesh, constitutional provision, role of BELA in the legal activism in Bangladesh, Bangladesh position on ICTPs; ECA 1995, ECR 1997, EC 2000 and other recent laws, Environmental policy: National and international.
4. Sectoral environmental legislation's in Bangladesh: laws regarding Biodiversity conservation, Forestry, Fishery, Mineral exploration, Groundwater management etc.
5. Relation of Environmental Laws to Ethics and ethical issues in environment. Role of classical and contemporary ethical theories to the suggestive measures for protecting environmental degradation both at local and global levels.

Recommended References:

1. Mohanty S.K., Universal's Environment & Pollution Law Manual, Universal Law publishing Co. Pvt. Ltd., 1999.
2. Laws Regulating the Environment in Bangladesh, M. Farooque and R. Hassan, published by BELA.
3. Rahman A.A., Haq S., Haither R., Jansen E.G. (eds), Environment & Development in
4. Bangladesh, voll 1&2, UPL, 1994.
5. Pojman P.L., Global Environmental Ethics, Mayfield Publishing Company, 2000
6. Rolston-3 H., Environmental Ethics, Temple University press, Philadelpfia, 1998.

Paper Code	244413	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Instrumental Techniques in Environmental Analysis**			

1. **Environmental Analysis and Laboratory Management:** Quality control and quality assurance, sample preparation and dissolution, chain of custody, compliance monitoring of

gas, liquid and solid samples, quantification of toxic metals in solid and liquid wastes, monitoring of trace elements in foods and biological samples.

2. Spectroscopy: The nature of radiant energy, interaction of light with matter, laws of photochemistry, quantum yields, wave mechanics, atomic and molecular absorption, origin of spectra, absorption and emission spectra.
3. UV-Visible Spectroscopy: Basic principles of spectroscopy, color detection, wavelength variation, absorption spectra of functional groups, instrumentation, absorption standards, stray correction, spectra-structure correlation, application.
4. Infrared Spectroscopy: Molecular motion, vibrational mode of molecules, working principle, sample techniques, spectral manipulation, baseline correction, transmission spectra of solid, liquid and gas phase, advantages of FT IR.
5. Atomic Absorption Spectroscopy: Theory of AAS, atomization techniques, methods of calibration, standard addition, scope, sensitivity and application of ICP-AAS, ICP-AES, ICP-MS techniques in environmental analysis.
6. Application of NMR and Mass Spectroscopy: principle and application of NMR, application of NMR to other nuclei, Mass spectra, ion sources, Mass spectrometer, application of molecular mass spectrometry, environmental application of NMR and mass spectrometry.
7. **Chromatographic Method: Principles of chromatography, instrumentation and applications of chromatography, chromatographic optimization, matrix preparation, HPLC in biological and forensic analysis, chromatographic separation of DDT and PCBs.**
8. **Other techniques: Thermogravimetric methods (TG), X-ray Emission, X-ray fluorescence and electroanalytical techniques; potentiometry, coulometry.**

Recommended References:

1. Spectroscopic Method in Organic Chemistry, D.H. Williams, I. Fleming, 4th Edn, Tata McGraw-Hill, 1988.
2. UV Spectroscopy, Techniques, Instrumentation, Data Handling, B.J. Clark, I. Frost, M.A. Russell, Chapman & Hall, 1993.
3. High Performance Liquid Chromatography, Fundamental, Principles and Practice, W.J. Lough, I.W. Wainer (Edn), Blackie Academic & Professional, 1991.
4. Understanding Instrumental Analysis, J.W. Robinson, 5th Edn, Marcel Decker, Inc., 1995.
5. Atomic Absorption Spectroscopy, Theory, Design & Application, S.D. Haswell (Edn), Elsevier, Vol 5, 1991.
6. Infrared Spectral Interpretation: A Systemic Approach, B. Smith, CRC Press, 1999.
7. Essential Environmental Science: Methods and Techniques by Simon Watts

Paper Code	244415	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	Energy, Environment and Sustainable Development: Bangladesh Perspective			

1. Energy: Definition, forms and classification of energy; Units of energy; Energy usage in past civilisations; role of energy use in the evolution of human civilization; Energy scenario and utilization pattern in Bangladesh.

2. Renewable energy sources (solar energy, hydropower, wind energy, bio-mass, bio-gas, tidal, geo-thermal etc.); advantages and disadvantages of renewable energy; future prospects.
3. Energy use and Environmental Pollution; Impacts of fossil fuel burning at local, regional and global levels; air pollution, climate change, global warming etc.
4. Energy conservation and Efficiency: Energy conversion process and Efficiency; importance of energy conservation, techniques of energy saving in house hold, industrial and transport sector; energy saving initiatives in Bangladesh; improved stove, energy saving bulb, cogeneration etc. and National energy policy of Bangladesh.
5. Energy and Sustainable Development: Concept of sustainable development; Energy resources and estimation of energy reserves and resources: Energy security and crisis Demand for energy services; Energy trade patterns and globalization; Energy prices and taxes; Energy investment; Global and regional energy needs.
6. Economic and Social Issues for Sustainable Energy Development: Energy and economic prosperity: energy consumption and economic well being; Disparities in Income and energy consumption, Energy intensities.
7. Energy and socio-political issues: Energy and poverty Energy and women Energy and population Energy and urbanization and Energy and life styles;
8. Energy, Environment and Health: Rural energy in the developing countries and impacts; Energy environment and health issues in house hold, workplace, community.
9. Energy and Geopolitical Issues: Politics regarding fossil fuel (oil, coal, gas) at National, Regional and International regarding.

Recommended References:

1. Joseph Priest, Energy Principles, Problems and Alternatives, 3rd Edition, Addision-Wesley Publishing Company.
2. Devid Reed, Editor, Structural Adjustment, the Environment, and Sustainable Development, Earthscan Eastern Publications Ltd. London.
3. A. K. Srivastava, Sustainable Development. Dragana Pilipovic, Energy Risk, McGraw-Hill.

Paper Code	244417	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Agriculture and Environment**			

1. **Historical perspective of agriculture:** Environmental factors and adaptation of agricultural plants: Basic principles of agricultural crop production, Concept of agroforestry and management, Soil, land and cropping pattern in Bangladesh, agro-ecological zones in Bangladesh.
2. **Agro-meteorology:** Introduction, objectives, data sources, estimation crop yields, How to monitor and use meteorological data for agriculture purpose, speculation of future trend of agricultural changes.
3. **Agrochemical and Environment:** Historical perspectives of agrochemical uses, Basic principle of agriculture crop production, Natural hazards, types of agrochemicals, Effect of agrochemicals on soil, Fresh water and plant kingdom, Agrochemical residues, misuse of

agrochemicals, Pollution Control in Agriculture: Slow release fertilizer, Integrated Pest Management (IPM).

4. **Impact of climate changes on agricultural production:** Bangladesh and global perspectives, understanding the driving forces of land use changes in Bangladesh, option for mitigation of the impact of climate changes, environmental changes and food security of Bangladesh and other densely populated countries.
5. **Natural problems in agriculture practices:** flood, drought, cyclone, rainfall, salinization, pest and disease, mitigation of natural problems.
6. **Agricultural practice and technology:** Choice of technology and technology transfer: import of environmental friendly technology from other Asian countries, Sustainable agricultural development in Bangladesh.
7. **Production of transgenic plants:** micropropagation, secondary metabolites production, impact of genetically modified food and biosafety.

Recommended References:

1. Mishra SJ and Mani D (1994) Agricultural pollution. Ashish Pub, New Delhi.
2. Whyte RO (1980) Crop Production and Environment. Faber and Faber Ltd, London.
3. Hugh Brammer (1999) Agricultural disaster Management in Bangladesh, UPL, Dhaka.
4. Chawla H.S. (2002) Introduction to plant biotechnology, 2nd Ed. Oxford & IBH publishing Company

Paper Code	**244418**	**Marks: 100**	**Credits: 4**	
Paper Title:	**Viva voce**			